

SAT UZBEKISTAN

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1.

Task	Time (minutes)
A	8
B	6
C	14
D	11
E	11

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

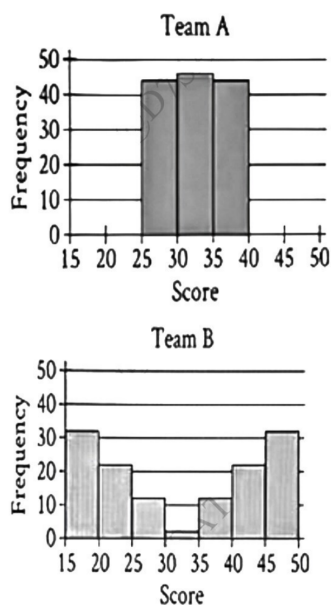
(A) 8

(B) 10

(C) 11

(D) 14

2.



The histograms summarize the distributions of scores for team A and team B. Which statement best compares the standard deviations of scores for these teams?

- ☐ A The standard deviation of scores for team A is less than the standard deviation of scores for team B.
- ☐ B The standard deviation of scores for team A is greater than the standard deviation of scores for team B.
- ☐ C The standard deviation of scores for team A is equal to the standard deviation of scores for team B.
- ☐ D There is not enough information to compare the standard deviations.

3.

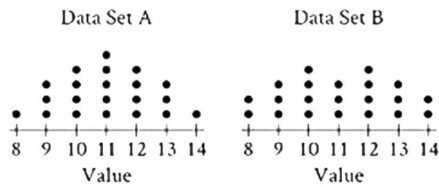
The list gives the mass, in grams, of 5 ruffed lemurs.

3,810; 3,810; 3,030; 3,850; 3,050

What is the mean mass, in grams, of these 5 ruffed lemurs?

4.

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.
- II. The standard deviation of data set A is equal to the standard deviation of data set B.

- ☐ (A) I and II
- ☐ (B) I only
- ☐ (C) II only
- ☐ (D) Neither I nor II

5.

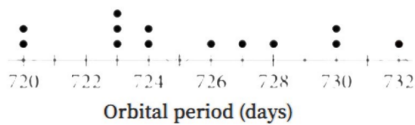
A company that produces only one type of item wants to estimate the percent of the items produced in a typical week that are defective. A random sample of 400 of the items produced in a certain week were tested. Based on the sample, it is estimated that 8.5% of all items produced by the company in this week are defective, with an associated margin of error of 2.73%. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all items produced by the company during this week?

- ☐ (A) 2.73% of the items are defective.
- ☐ (B) It is plausible that between 5.77% and 11.23% of the items are defective.
- ☐ (C) 8.5% of the items are defective.
- ☐ (D) It is plausible that more than 11.23% of the items are defective.

6.

The list gives the mass, in grams, of 5 ruffed lemurs.
3,910; 3,910; 3,130; 3,950; 3,150
What is the mean mass, in grams, of these 5 ruffed lemurs?

7.



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 632 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A

The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.
- B

The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- C

The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D

The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

8.

The list gives the number of individuals in each of 6 groups of ring-tailed lemurs.
17, 4, 20, 17, 18, 6

What is the range of the numbers of individuals for the 6 groups of ring-tailed lemurs?

9.

Task	Time (minutes)
A	6
B	4
C	9
D	8
E	8

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

- A

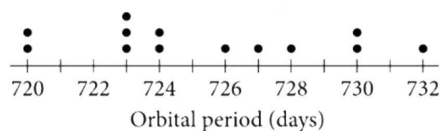
5
- B

7
- C

8
- D

9

10.



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 251 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- ☐ A The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.
- ☐ B The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- ☐ C The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- ☐ D The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

11.

A company that produces only one type of item wants to estimate the percent of the items produced in a typical week that are defective. A random sample of 540 of the items produced in a certain week were tested. Based on the sample, it is estimated that 12.5% of all items produced by the company in this week are defective, with an associated margin of error of 2.79%. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all items produced by the company during this week?

- ☐ A 2.79% of the items are defective
- ☐ B It is plausible that between 9.71% and 15.29% of the items are defective
- ☐ C 12.5% of the items are defective
- ☐ D It is plausible that more than 15.29% of the items are defective

12.

Ari examined a set of 86 plants. The lightest plant had a mass of 2.8 kilograms (kg) and the heaviest plant had a mass of 6.2 kg. Chihiro examined the same set of 86 plants and also an additional plant with a mass of 11.7 kg. Which of the following must be true about the set of plants that Ari examined and the set of plants that Chihiro examined?

- ☐ A The standard deviation of the masses, in kg, of the plants that Ari examined is greater than the standard deviation of the masses, in kg, of the plants that Chihiro examined.
- ☐ B The range of the masses, in kg, of the plants that Ari examined is greater than the range of the masses, in kg, of the plants that Chihiro examined.
- ☐ C The median of the masses, in kg, of the plants that Ari examined is less than the median of the masses, in kg, of the plants that Chihiro examined.
- ☐ D The mean of the masses, in kg, of the plants that Ari examined is less than the mean of the masses, in kg, of the plants that Chihiro examined.

13.

Data set A: 8, 11, 12, 15, 21, 27
 Data set B: 9, 12, 13, 16, 22, h

Data sets A and B each consist of 6 values as shown, where h is a constant. If the standard deviation of data set A is greater than the standard deviation of data set B, which of the following could be the value of h ?

I. 23
 II. 28
 III. 29

- ☐ A I only
- ☐ B II only
- ☐ C III only
- ☐ D II or III

14.

A list of 10 data values is shown.
 10, 14, 22, 6, 24, 26, 14, 8, 8, 8

What is the mean of these data?

- ☐ A 8
- ☐ B 12
- ☐ C 14
- ☐ D 20

15.

Task	Time (minutes)
A	8
B	5
C	12
D	10
E	10

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

- ☐ A 7
- ☐ B 9
- ☐ C 10
- ☐ D 12

16.

Weight (pounds)	Frequency
13	12
14	10
15	3
16	5
17	7
18	12
19	13
20	9

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of 71 *Ardeotis kori* birds. A weight of 32 pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of 72 *Ardeotis kori* birds. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- ☐ A The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.
- ☐ B The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
- ☐ C The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original
- ☐ D The mean of the new data set is less than the mean of the original data set, and the median of the two data sets are equal.

17.

Ari examined a set of 85 plants. The lightest plant had a mass of 2.8 kilograms kg and the heaviest plant had a mass of 6.2 kg. Chihiro examined the same set of 85 plants and also an additional plant with a mass of 11.6 kg. Which of the following must be true about the set of plants that Ari examined and the set of plants that Chihiro examined?

- ☐ A The standard deviation of the masses, in kg, of the plants that Ari examined is greater than the standard deviation of the masses, in kg, of the plants that Chihiro examined.
- ☐ B The range of the masses, in kg, of the plants that Ari examined is greater than the range of the masses, in kg, of the plants that Chihiro examined.
- ☐ C The median of the masses, in kg, of the plants that Ari examined is less than the median of the masses, in kg, of the plants that Chihiro examined.
- ☐ D The mean of the masses, in kg, of the plants that Ari examined is less than the mean of the masses, in kg, of the plants that Chihiro examined.

18.

Which of the following lists represents a data set with the smallest standard deviation?

- (A) 44,45,47,49,50
- (B) 45,45,47,49,49
- (C) 45,46,47,48,49
- (D) 46,47,47,47,48

19.

Mass(grams)	Frequency
10	12
20	6
30	8
40	9

The frequency table summarizes the mass, in grams, of 35 objects. What is the mean mass, in grams, of these objects?

Answer Preview:

20.

Data set A: 24, 34, 44, 54, 64

Data set A, which consists of the 5 values shown, has a mean of 44 and a standard deviation of approximately 14.10. Data set B consists of the 5 values from data set A and one additional value, k . For which of the following values of k will the standard deviation of data set B be less than the standard deviation of data set A?

- (A) 14
- (B) 49
- (C) 74
- (D) There is no value of k such that the standard deviation of data set B will be less than the standard deviation of data set A.

21.

Values	Data set A frequency	Data set B frequency	Data set C frequency	Data set D frequency
40	0	0	8	8
43	2	3	3	4
46	4	3	3	2
49	8	8	0	0

The table shows the frequencies of the data values for four data sets. Which data set has the greatest mean?

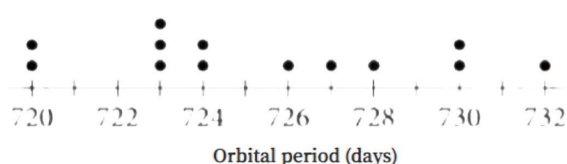
(A) Data set A

(B) Data set B

(C) Data set C

(D) Data set D

22.



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 621 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

(A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

(B) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

23.

A district school board in a certain state is proposing a change to the length of lunch periods for all high schools in the district. A sample of 365 high school students was selected at random from all high school students in the district. The selected students were asked whether they approved of the proposed change, and 211 students responded that they did not approve. Which of the following is the largest population to which the results of the survey can be generalized?

- (A) All high school students in the district
- (B) The 365 students who were surveyed
- (C) All high school students in the state
- (D) The 211 students who responded that they did not approve of the proposed change

24.

x	y
75	11
80	19
85	6
90	5
95	10
Total	51

Ari will add a 135-year-old stamp to her original collection, creating a new collection of 52 stamps. Which statement correctly compares the mean ages and median ages of Ari's original and new collections?

- (A) The mean age for the new collection is greater than the mean age for the original collection, and the median age for the new collection is greater than the median age for the original collection.
- (B) The mean age for the new collection is greater than the mean age for the original collection, but the median age is the same for the original and new collections.
- (C) The mean age for the new collection is less than the mean age for the original collection, and the median age for the new collection is less than the median age for the original collection.
- (D) The median age for the new collection is greater than the median age for the original collection, but the mean age is the same for the original and new collections.

25.

Data set A: 2, 3, 4, 5, 5
Data set B: 2, 3, 4, 10

Which of the following statements best compares the means of data set A and data set B?

- (A) The mean of data set A is less than the mean of data set B.
- (B) The mean of data set A is greater than the mean of data set B.
- (C) The means of the two data sets are equal.
- (D) There is not enough information to compare the means.

26.

The values in data sets X and Y are shown in the table.

Data set X	13	13	14	14	15	16	16	17	17
data set Y	2	2	3	3	4	5	5	6	6

The standard deviation of data set X is q , and the standard deviation of data set Y is s . Which of the following statements about the standard deviation of the data sets is true?

- (A) $q < s$
- (B) $q > s$
- (C) $q = s$
- (D) The relationship between q and s cannot be determined.

27.

The table shows the distribution of height, in centimeters, of 25 objects.

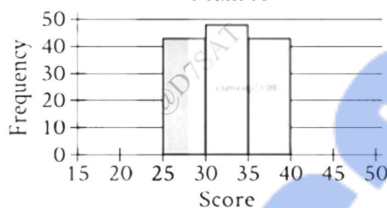
Height	Frequency
0 to 100	8
101 to 200	6
201 to 300	5
301 to 400	4
401 to 500	2

Which of the following could be the median height of these objects?

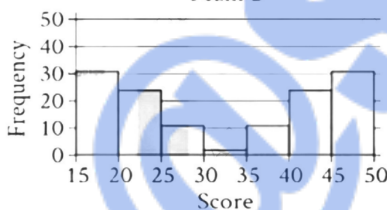
- (A) 355
- (B) 255
- (C) 155
- (D) 55

28.

Team A



Team B



The histograms summarize the distributions of scores for team A and team B. Which statement best compares the standard deviations of scores for these teams?

- (A) The standard deviation of scores for team A is less than the standard deviation of scores for team B.
- (B) The standard deviation of scores for team A is greater than the standard deviation of scores for team B.
- (C) The standard deviation of scores for team A is equal to the standard deviation of scores for team B.
- (D) There is not enough information to compare the standard deviations.

29.

Data Set X	
Value	Frequency
a	0
b	5
c	19

Data Set Y	
Value	Frequency
a	1,900
b	500
c	0

Data sets X and Y are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

- ☐ A The mean of data set X is $\frac{2(19)}{24}$ less than the mean of data set Y. 3or5
- ☐ B The mean of data set X is $\frac{2(19)}{24}$ greater than the mean of data set Y.
- ☐ C The mean of data set Y is 100 times the mean of data set X.
- ☐ D The means of the two data sets are equal.

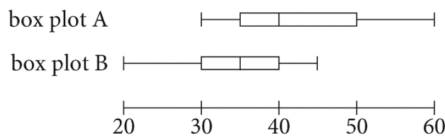
30.

27, 8, 20, 43, 32

What is the mean of the given data set?

- A) 20
B) 26
C) 27
D) 35

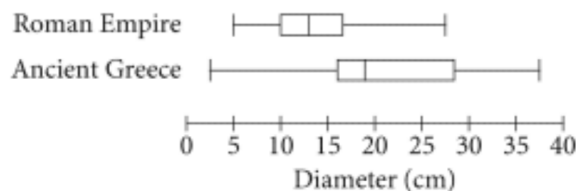
31.



Based on the two box plots above, which of the following statements is true?

- A) The range of box plot A is greater than the range of box plot B.
- B) The range of box plot A is less than the range of box plot B.
- C) The range of box plot A is equal to the range of box plot B.
- D) It cannot be determined if the range of box plot A is different from the range of box plot B.

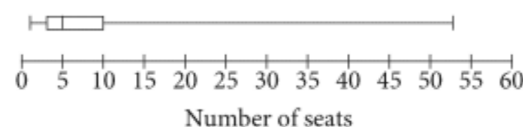
32.



The Metropolitan Museum of Art has plates on display from the Roman Empire and ancient Greece. The box plots shown summarize the distributions of the diameters, in centimeters, of all the museum's plates from each region. How does the median diameter of the plates from the Roman Empire, r , compare to the median diameter of the plates from ancient Greece, g ?

- A) $r < g$
- B) $r > g$
- C) $r = g$
- D) There is not enough information to compare the medians.

33.



The box plot summarizes the number of seats in the US House of Representatives currently allocated to each of the 50 states. What is the median number of allocated seats in the US House of Representatives?

- A) 2
- B) 5
- C) 10
- D) 53

34.

The dot plot shows the number of eggs in each of 13 wood duck nests.



How many of these nests had exactly 11 eggs?

- A) 1
- B) 2
- C) 3
- D) 4

35.

Data set A: 10, 10, 10, 10, 10, 200

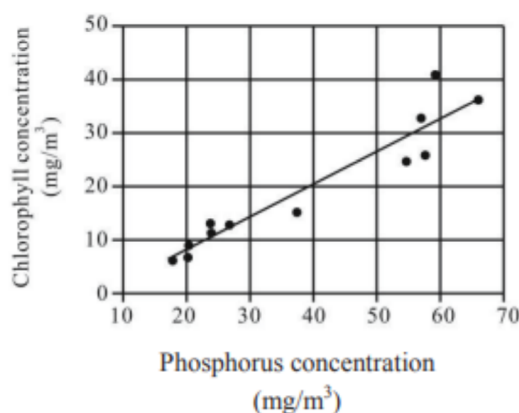
Data set B: 10, 10, 10, 10, 10

Which of the following statements correctly compares the mean of data set A and the mean of data set B?

- A) The mean of data set A is less than the mean of data set B.
- B) The mean of data set A is equal to the mean of data set B.
- C) The mean of data set A is greater than the mean of data set B.
- D) The relationship between the two means cannot be determined from the given information.

36.

Concentrations of Phosphorus and Chlorophyll in a Lake



The scatterplot above shows the concentrations, in milligrams per cubic meter (mg/m^3), of phosphorus and chlorophyll that were measured in a lake for 12 different years. A line of best fit for the data is also shown. According to the line of best fit, when the concentration of phosphorus in the lake is $40 \text{ mg}/\text{m}^3$, which of the following is the best prediction for the concentration of chlorophyll in the lake?

- A) $60 \text{ mg}/\text{m}^3$
- B) $40 \text{ mg}/\text{m}^3$
- C) $20 \text{ mg}/\text{m}^3$
- D) $10 \text{ mg}/\text{m}^3$

37.

2, 6, 10, x , 1, 9, 5

The median of the seven given numbers is 6. Which of the following could be the value of x ?

- A) 3
- B) 4
- C) 5
- D) 6

38.

In a desert region, the amount of sunlight falling on a surface with an area of one square meter is 1,000 watts. Which of the following could be the amount of sunlight, in watts, reflected from this one-square-meter surface?

- A) 275
- B) 525
- C) 725
- D) 1,275

39.

Darnell made a list of the amount of time, in minutes, he spent practicing the piano for each of the last 7 days. Darnell's list is shown.

21, 15, 22, 31, 35, 91, 18

After making the list, Darnell realized that the outlier of 91 minutes was an error, and he corrected it to 19 minutes. Which of the following best describes how the median time changed when the outlier of 91 minutes was corrected to 19 minutes?

- A) The median increased.
- B) The median decreased.
- C) The median stayed the same.
- D) There is not enough information to determine how the median changed.

40.

The list of numbers shown represents the ages of 6 different children.

4, 7, 5, 9, 7, 3

What is the median age?

- A) 5
- B) 6
- C) 7
- D) 9

41.

Data set X: 5.50, 5.50, 5.60, 5.65, 5.66

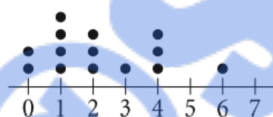
Data set Y: 4.00, 5.50, 5.50, 5.60, 5.65, 5.66

Data sets X and Y show the acidity, or pH, of rainwater samples from two different locations. Which statement about the mean pH of data set X and data set Y is true?

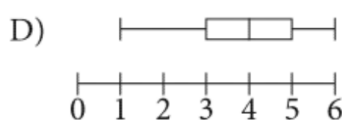
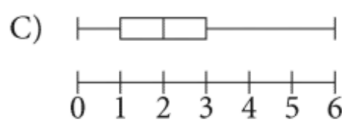
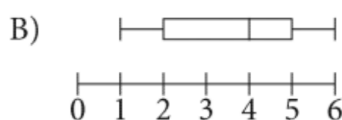
- A) The mean pH of data set X is greater than the mean pH of data set Y.
- B) The mean pH of data set X is less than the mean pH of data set Y.
- C) The mean pH of data set X is equal to the mean pH of data set Y.
- D) There is not enough information to compare the mean pH of the two data sets.

42.

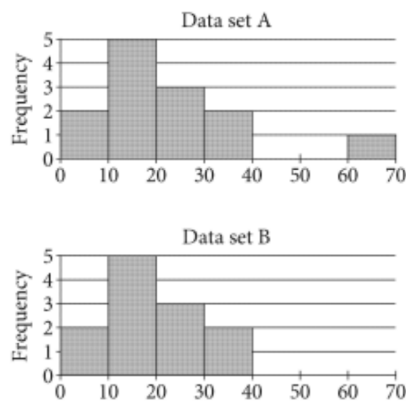
The dot plot below shows the number of days it rained in a certain week for 14 cities in the southeastern United States.



Which of the following box plots correctly represents these data?



43.



The two histograms show the distribution of data set A and data set B, respectively. Data set B is the result of removing the outlier from data set A. Which of the following statements about the means of data set A and data set B is true?

- A) The means of data set A and B are the same.
- B) The mean of data set A is greater than the mean of data set B.
- C) The mean of data set A is less than the mean of data set B.
- D) No comparison about the means of the data sets can be made.

44.

The list gives the diameter, in inches, of the five aspen trees at a farm.

14.4, 14.9, 14.9, 15.6, 16.2

Another aspen tree with a diameter of 10.6 inches is planted at the farm. How does the median diameter of the aspen trees at this farm change when the sixth tree is added?

- A) The median increases.
- B) The median decreases.
- C) The median does not change.
- D) There is not enough information to determine how the median changes.